Department of Computer Science – Mini Project 03

📜 Project Title:

Attendance Management System of Teachers

🧩 Problem Statement / Project Scenario:

A university wants to automate the attendance tracking of its faculty members. Each teacher's **in-time**, **out-time**, **total hours**, **late-ins**, **early-gos**, **short hours**, and **extra hours** should be tracked. Additionally, different types of staff (permanent, visiting, administrative) have different attendance policies. The system must efficiently use **inheritance** to reuse code, and incorporate **constructors, destructors, objects, copy constructors**, and various **inheritance models (single, multiple, multilevel, hierarchical, hybrid)** with different **access specifiers**.

1. Class: Person

- **Purpose**: Serves as the base class for all human entities in the system, such as teachers and admins.
- **Inheritance**: Base class (does not inherit from any class).
- **Access Specifiers & Data Members**:
  - protected:
    - string name: Name of the person.
    - int id: Unique ID of the person.
- **Member Functions**:
  - Person(): Default constructor.
  - Person(string n, int i): Parameterized constructor.
  - void getPersonalInfo(): Takes personal information from the user.
  - void displayPersonalInfo(): Displays stored personal information.
  - ~Person(): Destructor to clean up resources.

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2. Class: TimeRecord

- **Purpose**: Maintains check-in and check-out times of a person.
- **Inheritance**: Independent class (not part of the hierarchy).
- **Access Specifiers & Data Members**:
  - protected:
    - string inTime: Time when the teacher checks in.
    - string outTime: Time when the teacher checks out.
- **Member Functions**:
  - TimeRecord(): Default constructor.
  - void setInTime(string time): Set check-in time.
  - void setOutTime(string time): Set check-out time.

- string getInTime(): Retrieve check-in time.
- string getOutTime(): Retrieve check-out time.

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3. Class: Attendance

- **Purpose**: Records attendance details including late arrival, early leave, short hours, and excess hours.
- **Inheritance**: **Publicly inherits** from TimeRecord.
- **Access Specifiers & Data Members**:
  - protected:
    - int lateInMinutes
    - int earlyGoMinutes
    - int shortHours
    - int excessHours
- **Member Functions**:
  - Attendance(): Default constructor.
  - void calculateLateIn(): Computes how late the teacher checked in.
  - void calculateEarlyGo(): Computes early departure.
  - void calculateShortAndExcessHours(): Calculates and differentiates between shortfall and excess hours.
  - void displayAttendanceReport(): Displays the full attendance details.

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4. Class: Teacher

- **Purpose**: Represents a teacher in the institution and tracks attendance.
- **Inheritance**: **Publicly inherits** from both Person and Attendance (**Multiple Inheritance**).
- **Access Specifiers & Data Members**:
  - private:
    - string subject: Subject the teacher teaches.
    - string department: Department the teacher belongs to.
- **Member Functions**:
  - Teacher(): Default constructor.
  - Teacher(string n, int i, string sub, string dept): Parameterized constructor.
  - void getTeacherDetails(): Input teacher-specific details.
  - void displayTeacherInfo(): Display full teacher profile and attendance.

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5. Class: Admin

- **Purpose**: Administrator who can add, remove, or view teacher data and reports.
- **Inheritance**: **Publicly inherits** from Person (Single Inheritance).

- **Access Specifiers & Data Members**:
  - private:
    - string role: Role of admin (e.g., Head, Supervisor).
- **Member Functions**:
  - Admin(): Default constructor.
  - Admin(string n, int i, string r): Parameterized constructor.
  - void manageTeachers(): Placeholder for managing teacher records.
  - void viewReports(): View attendance summaries.

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6. Class: Department

- **Purpose**: Handles groupings of teachers by departments.
- **Inheritance**: **Protected Inheritance** from Teacher (Hierarchical Inheritance).
- **Access Specifiers & Data Members**:
  - private:
    - string departmentName
    - int numberOfTeachers
- **Member Functions**:
  - Department(): Default constructor.
  - void setDepartmentInfo(string name, int count): Set department info.
  - void displayDepartmentInfo(): Display department summary.

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7. **Class: ReportGenerator**

Purpose: Generates full attendance summaries and analytics. It compiles and analyzes teacher attendance records, highlighting key metrics such as late-ins, early-outs, short hours, and excess hours.

Inheritance: Publicly inherits from `Teacher` (can also be considered hybrid if further extended).

Access Specifiers & Data Members:

- **private**:
  - ♣ `Teacher t1, t2, t3;` — Stores a fixed number of teacher objects for reporting.
  - ♣ `int teacherCount;` — Maintains how many teachers are being considered.
  - ♣ `string reportDate;` — Stores the date of the report generation.

Member Functions:

- `ReportGenerator()` — Constructor to initialize report generator and default teacher values.
- `void inputTeachers()` — Takes input for teachers' attendance and personal data.
- `void generateDailyReport()` — Prints attendance information for the current day for all teachers.
- `void generateMonthlySummary()` — Summarizes and displays cumulative attendance statistics over a month.

o `void generateUserSelectedDateSummary()` — Summarizes and displays cumulative attendance statistics between selected range of dates.

o `~ReportGenerator()` — Destructor to clean up any used resources or reset values.